

# A Practical Guide to Medical image computing

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TOPIC -- MEDICAL IMAGE COMPUTING

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Data forms Medical image computing typically operates on uniformly sampled data with regular x-y-z spatial spacing (images in 2D and volumes in 3D, generically referred to as images). At each sample point, data is commonly represented in integral form such as signed and unsigned short (16-bit), although forms from unsigned char (8-bit) to 32-bit float are not uncommon. The particular meaning of the data at the sample point depends on modality: for example a CT acquisition collects radiodensity values, while an MRI acquisition may collect T1 or T2-weighted images. Longitudinal, time-varying acquisitions may or may not acquire images with regular time steps. Fan-like images due to modalities such as curved-array ultrasound are also common and require different representational and algorithmic techniques to process. Other data forms include sheared images due to gantry tilt during acquisition; and unstructured meshes, such as hexahedral and tetrahedral forms, which are used in advanced biomechanical analysis (e.g., tissue deformation, vascular transport, bone implants).

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Software Software for medical image computing is a complex combination of systems providing IO, visualization and interaction, user interface, data management and computation. Typically system architectures are layered to serve algorithm developers, application developers, and users. The bottom layers are often libraries and/or toolkits which provide base computational capabilities; while the top layers are specialized applications which address specific medical problems, diseases, or body systems.

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Multiple templates Rather than relying on a single template, multiple templates can be used. The idea is to represent an image as a deformed version of one of the templates. For example, there could be one template for a healthy population and one template for a diseased population. However, in many applications, it is not clear how many templates are needed. A simple albeit computationally expensive way to deal with this is to have every image in a training dataset be a template image and thus every new image encountered is compared against every image in the training dataset. A more recent approach automatically finds

the number of templates needed.

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Massive univariate approaches that probe individual voxels in the imaging data for a relationship to the experiment condition. The prime approach is the general linear model. Multivariate- and classifier based approaches, often referred to as multi voxel pattern analysis or multi-variate pattern analysis probe the data for global and potentially distributed responses to an experimental condition. Early approaches used support vector machines to study responses to visual stimuli. Recently, alternative pattern recognition algorithms have been explored, such as random forest based gini contrast or sparse regression and dictionary learning. Functional connectivity analysis studies the intrinsic network structure of the brain, including the interactions between regions. The majority of such studies focus on resting state data to parcelate the brain or to find correlates to behavioral measures. Task specific data can be used to study causal relationships among brain regions (e.g., dynamic causal mapping). When working with large cohorts of subjects, the normalization (registration) of individual subjects into a common reference frame is crucial. A body of work and tools exist to perform normalization based on anatomy (FSL, FreeSurfer, SPM). Alignment taking spatial variability across subjects into account is a more recent line of work. Examples are the alignment of the cortex based on fMRI signal correlation, the alignment based on the global functional connectivity structure both in task-, or resting state data, and the alignment based on stimulus specific activation profiles of individual voxels.

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Segmentation is the process of partitioning an image into different meaningful segments. In medical imaging, these segments often correspond to different tissue classes, organs, pathologies, or other biologically relevant structures. Medical image segmentation is made difficult by low contrast, noise, and other imaging ambiguities. Although there are many computer vision techniques for image segmentation, some have been adapted specifically for medical image computing. Below is a sampling of techniques within this field; the implementation relies on the expertise that clinicians can provide.

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Journals on medical image computing Medical Image Analysis (MedIA) ; also the official journal of The MICCAI Society, which organizes the Annual MICCAI Conference a premier conference for medical image computing IEEE Transactions on Medical Imaging (IEEE TMI) Medical Physics Journal of Digital

Imaging (JDI) ; the official journal of the Society of Imaging Informatics  
Computerized Medical Imaging and Graphics Journal of Computer Aided  
Radiology and Surgery BMC Medical Imaging In addition the following journals  
occasionally publish articles describing methods and specific clinical applications  
of medical image computing or modality specific medical image computing

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Estimation of local tissue properties, such as diffusivity; Estimation of local directions and global pathways of diffusion. The diffusion tensor, a  $3 \times 3$  symmetric positive-definite matrix, offers a straightforward solution to both of these goals. It is proportional to the covariance matrix of a Normally distributed local diffusion profile and, thus, the dominant eigenvector of this matrix is the principal direction of local diffusion. Due to the simplicity of this model, a maximum likelihood estimate of the diffusion tensor can be found by simply solving a system of linear equations at each location independently. However, as the volume is assumed to contain contiguous tissue fibers, it may be preferable to estimate the volume of diffusion tensors in its entirety by imposing regularity conditions on the underlying field of tensors. Scalar values can be extracted from the diffusion tensor, such as the fractional anisotropy, mean, axial and radial diffusivities, which indirectly measure tissue properties such as the dysmyelination of axonal fibers or the presence of edema. Standard scalar image computing methods, such as registration and segmentation, can be applied directly to volumes of such scalar values. However, to fully exploit the information in the diffusion tensor, these methods have been adapted to account for tensor valued volumes when performing registration and segmentation. Given the principal direction of diffusion at each location in the volume, it is possible to estimate the global pathways of diffusion through a process known as tractography. However, due to the relatively low resolution of diffusion MRI, many of these pathways may cross, kiss or fan at a single location. In this situation, the single principal direction of the diffusion tensor is not an appropriate model for the local diffusion distribution. The most common solution to this problem is to estimate multiple directions of local diffusion using more complex models. These include mixtures of diffusion tensors, Q-ball imaging, diffusion spectrum imaging and fiber orientation distribution functions, which typically require HARDI acquisition with a large number of gradient directions. As with the diffusion tensor, volumes valued with these complex models require special treatment when applying image computing methods, such as registration and segmentation.